William Theodor Lehn-Schiøler

Industrial PhD Student · BrainCapture & DTU Health Tech

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Researcher building large EEG foundation models for clinical use. My work combines self-supervised learning and explainable AI to turn neural representations into transparent, actionable insights for clinicians — with the aim of automating EEG interpretation and making neurological diagnostics available in resource-limited settings.

EXPERIENCE

BrainCapture, Lyngby — *Industrial PhD Student* SEP 2025 – PRESENT

In collaboration with DTU Health Tech (Digital Health). Developing large-scale EEG foundation models and methods to interpret them — including sparse-autoencoder–based mechanistic interpretability, cross-domain pretraining on sleep data, and point-of-care tools such as real-time electrode detection. Co-supervising BSc/MSc theses and special courses at DTU.

BrainCapture, Lyngby — *Data Scientist* JAN 2021 – AUG 2025

- Led the design and development of **large-scale machine learning systems** for EEG analysis in **Python**, aimed at bridging diagnostic gaps in resource-limited areas.
- Drove **research-to-production integration**, translating ML prototypes into robust, optimised features in the core product.
- Applied **explainable AI** (concept-based explainability) to improve model transparency on medical-device data.
- Contributed to clinical evaluation of low-cost EEG, including field studies in Kenya.

DIS – Study Abroad in Scandinavia, Copenhagen — *External Lecturer* JAN 2024 – JUN 2024

DTU Compute, Lyngby — *Teaching Assistant* FEB 2022 – DEC 2023

02463 Active Machine Learning, 02464 Human Cognition and AI, 02471 Machine Learning for Signal Processing — communicating technical material clearly to undergraduate and graduate students.

EDUCATION

Technical University of Denmark — *PhD, DTU Health Tech (Digital Health)* SEP 2025 – PRESENT

Industrial PhD in collaboration with BrainCapture — EEG foundation models and interpretability.

Technical University of Denmark — *MSc Mathematical Modelling and Computing* SEP 2022 – DEC 2024

École Polytechnique – Université Paris-Saclay — *Exchange semester* SEP 2021 – JAN 2022

Double Majeure Mathématiques & Informatique

Technical University of Denmark — *BSc Artificial Intelligence and Data* SEP 2019 – JUN 2022

VOLUNTEER EXPERIENCE

Technical University Hospital, Copenhagen — *Student Ambassador* APR 2026 – PRESENT

Copenhagen MedTech, Copenhagen — *Board Member & Chairman* SEP 2022 – DEC 2024

PUBLICATIONS

Full and up-to-date list on Google Scholar and willsth.github.io/publications.

Preprints & under review

Real-Time EEG Cap Electrode Detection for Guided Point-of-Care Placement 2026
W Lehn-Schiøler, MS Nilsson, NS Detlefsen
arXiv:2607.20142

Clinical Feasibility of Smartphone-based EEG in Kenya 2026
W Lehn-Schiøler, N Enkhtsetseg, AM Storgaard, MG Pedersen, D Rice, G Wambugu, NJ Fidele, M Cacic Hribljan, AA Arbune, S Armand Larsen, S Beniczky
arXiv:2605.08157

Pretraining on Sleep Data Improves non-Sleep Biosignal Tasks 2026
W Lehn-Schiøler, MR Kjær, P Hempel, MG Pedersen, R Thapa, B He, N Spicher, A Brink-Kjær, LK Hansen, E Mignot
arXiv:2605.02500

Peer-reviewed

Mechanistic Interpretability of EEG Foundation Models via Sparse Autoencoders 2026
4 cit.
W Lehn-Schiøler, MR Kjær, R Thapa, MG Pedersen, AM Storgaard, N Williams, R Gatej, T Lehn-Schiøler, A Brink-Kjær, S Puthusserypady, S Beniczky, J Zou, LK Hansen
NeurIPS 2026 (to appear)

Feasibility of a Smartphone-based EEG Recording System for Neurological Evaluation in Kenya 2026
N Enkhtsetseg, **W Lehn-Schiøler**, NJ Fidele, T Lehn-Schiøler, et al.
Neurology 106 (11_Suppl. 1), AAN Annual Meeting (abstract)

Low-cost portable EEG device for bridging the diagnostic gap in resource-limited areas 2024
25 cit.
S Armand Larsen, L Klok, **W Lehn-Schiøler**, R Gatej, S Beniczky
Epileptic Disorders 26 (5), 694–700

SPEED: Scalable Preprocessing of EEG Data for Self-Supervised Learning 2024
4 cit.
A Gjølbye, L Skerath, **W Lehn-Schiøler**, N Langer, LK Hansen
IEEE MLSP 2024

Concept-based explainability for an EEG transformer model 2023
16 cit.
AG Madsen, **W Lehn-Schiøler**, Á Jónsdóttir, B Arnardóttir, LK Hansen
IEEE MLSP 2023